

AI and its use for energy facilities and urban planning.

Integrating AI into the US energy and environmental sector is a complex issue, with both policy makers and the public weighing up its potential benefits and significant ethical and practical challenges. While the use of AI streamlines processes, reduces timelines, and promises enhanced security, concerns over its energy use, lack of human judgement and potential biases are key to the discussion of its use case.

Environmental Permitting and Interconnection Processes

The DOE aims to invest \$30 million to use AI to accelerate the lengthy interconnection queues, working with grid operators and software developers to provide faster data analysis and application(DOE, 2025). However, some policy makers and environmental advocates express concerns over AI driven fast tracking, as it could lead to reduced public participation and opinion as well as a less rigorous environmental analysis, potentially leading to continuous oversight of unique case by case issues(Farber, 2025). With concerns over environmental protection and transparency, many think that AI driven permits may limit avenues for community discussion and overlook local impacts(ebit). AI falls short when it comes to environmental permitting. This is because of the bias that such algorithms fall into. As AI is trained with historical data, bias in the past is often carried forward or even magnified into its solutions in the present. These biases are difficult to remove and prevent AI from changing over time, as human policy makers would(Sustainability Directory, 2025). There is also a lack of transparency and accountability with using AI. Many AI systems operate as “Black Box’s”, providing solutions without explanation or reasoning, leading to reduced confidence and making scrutinizing or appealing AI generated outcomes difficult(Silva, 2023).

AI’s use for Zoning Analysis

AI is only as good as the information it has access to and can often return inadequate results due to imperfect information. For example, when prompted to verify the amount of residents in the New York county of Central Islip that are in K-12, the answer of 7569, is supported by data from the county school district. However, it is important to note that information regarding residents attending private or charter schools is not publicly available and hence an AI bot is unable to create a full and in depth assessment(Looney et al, 2025). This example shows that while AI is useful, it often fails to account for what it doesn't know and can create false conclusions because of this. The company Leafpress uses AI to analyze zoning codes and community sentiment to flag development issues early, while also showing areas of less strict policy(Leafpress et al, 2025). While this is useful for developers and their efficiency, community groups see it as a tool to bypass local opposition. In general many desire and champion AI for efficient energy development and decarbonization, such as more efficient trucking solutions and water use

optimization. Along with AI, cities must utilize the Internet of Things in order to capture enough data for predictive AI models to work. Cities such as Singapore and Barcelona have trialed programs which utilise extra sensors, GPS trackers and cameras in order to provide enough data for AI models to predict traffic and waste management levels and can change traffic light patterns and reroute waste trucks in order to increase flow and reduce wasted idling time(Cina et al, 2025). In Singapore, developers also use this data to make informed decisions on new urban developments. One such tool is a 3D model of the city which planners use to simulate new infrastructure before it is implemented(ebit), enabling planners to spot issues such as ventilation in dense areas and transport needs for residential to business sectors. The city of Singapore was able to develop these AI models due to favourable conditions in the AI space. Increased inter-agency collaboration and shared data along with the government's role in setting national standards in data governance resulted in reduced institutional resistance to AI adoption. One of the main reasons for mass scale AI adoption was due to citizen engagement and transparency, as trust in AI tools, such as surveillance and behavioral nudging, was strongly linked to the degree of public engagement in the design and rollout of such models(ebit).

Cyber threats to energy facilities

AI is a vital tool for safeguarding critical energy infrastructure, as cyber attacks become increasingly sophisticated, so too do the tools used to prevent them. New legislation introduced by 2 U.S. senators, Jim Risch R-Idaho and John Hickenlooper D-Colorado, aims to coordinate efforts by the DOE and the Cybersecurity and Infrastructure Security Agency to better assess and mitigate cyber threats(Ribeiro, 2025). Many industry leaders support the use of AI in cyber security, with 84% of decision makers and influences in the energy sector committing to current or future AI adoption in cybersecurity(Ribeiro, 2025)as well as in their day to day business. These industry leaders expressed that the most valuable aspects of AI were in cybersecurity and threat detection, predictive maintenance, and operational efficiency(ebit). This showcases the positive outlook that the industry has on the use of AI. However, especially in the security field, trust plays an important role in the adoption of AI, A study by KPMG showed that trust is directly correlated with the level of use of AI(KPMG, 2024), with many people using AI for its technical ability and accuracy rather than its ability to protect data or determine ethical outcomes such as security measures and cut offs(ebit). Again, AI runs into the issue of accountability, with automated responses potentially having unintended consequences such as grid shutdowns, if not absolutely necessary there is no one individual to take the blame and the issue is simply overlooked(Shark, 2025). AI may also overlook novel threats or attacks it hasn't been trained to look for, maintaining the need for human engineers to overlook the operations. In contrast, AI is useful for its ability to be active 24/7 and removes most human error or shortage(Dunn, 2025), allowing for better quality of life for security engineers and reduced day to day strain.

Additionally AI acceptance is lower in advanced economies, likely due to a greater trust of professionals and hence a reduced need for an AI helper(KPMG, 2024).

Conclusion

Overall, the recurring notion is that, currently, AI takes up too much energy and water when used and that most of the benefits gained from efficiency and speed are canceled out. Recent polls indicate that 72% of Americans are concerned about AI's increasing energy use with 61% supporting a tax on electricity used by AI datacentres to fund grid upgrades(Colson, 2025). Many also worry that rising energy usage could lead to higher electricity rates for consumers(Oberhaus, 2025). With governments investing heavily in AI systems such as the DOE's PermitAI and the development of smart cities such as Singapore and Barcelona, it is clear that AI adoption will continue to increase and develop over the coming years. With rapid increases in the amount and scale of new energy projects, interconnection backlogs have only increased in recent years which is why the DOE aims to utilize AI to "involve everyone across the interconnection system to help reimagine, retool, and shape the grid of the future"(DOE, 2025). While AI tools are championed for efficiency and speed, concerns over transparency and accountability are common causes for reduced adoption, especially when it comes to influencing human behaviors and urban development(Cina et al, 2025). As more and more human based decision making is replaced by AI, developers must design models to be transparent as "Black-box algorithms, which lack transparency in their decision-making processes,can lead to mistrust and suspicion"(Silva, 2023). In the cybersecurity space, the adoption of AI is crucial in order to stop and prevent increasingly sophisticated attacks on energy infrastructure. This is backed by members of congress such as Jim Risch R-Idaho, stating that an "Increased risk of cyberattacks requires more diligent information sharing to effectively monitor and mitigate threats to America's energy sector" and that "Idaho is already leading the way in combatting cyber threats through the Idaho National Lab. My Energy Threat Analysis Program Act will support these efforts and better protect the U.S. from future cyberattacks."(Ribeiro, 2025). With AI's ability to scan mass amounts of data and spot issues much faster than its human equivalent would, its use as a cybersecurity tool is invaluable and likely will be adopted to some degree by most governments and organizations across the world.

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